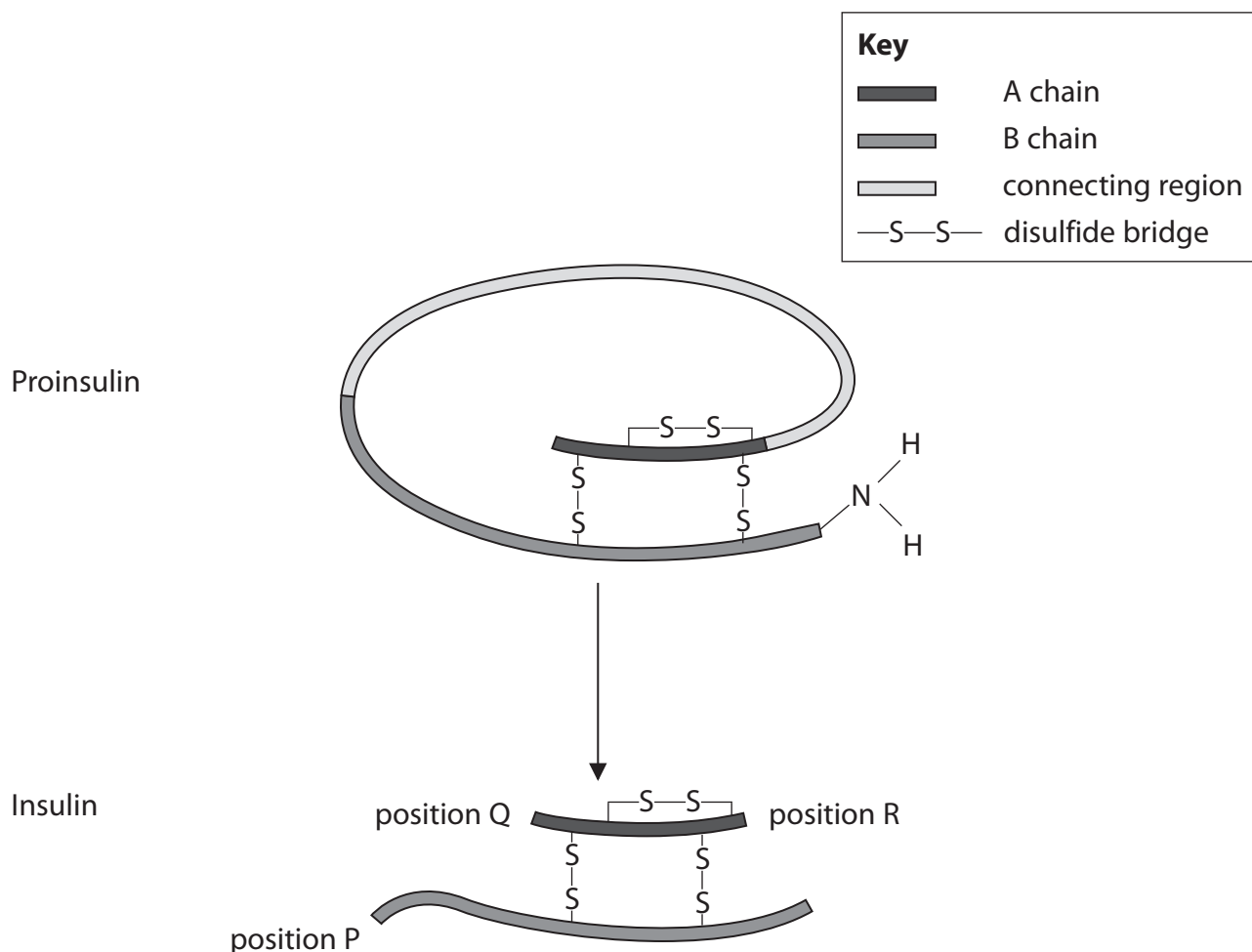


- 7 Insulin is a small protein composed of two peptide chains, the A chain and the B chain.

Insulin is made from the protein proinsulin.

Proinsulin is composed of the A chain and the B chain joined by a sequence of amino acids called the connecting region.

The diagram shows how insulin is formed from proinsulin.



- (a) The connecting region is removed from proinsulin, leaving the A chain connected to the B chain by disulfide bridges (—S—S—).

Which row of the table shows the position of an amino group (NH_2) and a carboxyl group (COOH)?

(1)

	Amino group	Carboxyl group
☐ A	position R	position P
☐ B	position P	position R
☐ C	position Q	position R
☐ D	position Q	position P

(b) Proinsulin is composed of 84 amino acids.

Insulin is composed of 51 amino acids.

(i) What percentage of proinsulin is the connecting region?

(1)

Answer%

(ii) There are three disulfide bridges in insulin. Each disulfide bridge is formed between the R groups of two cysteine amino acids.

Give the ratio of amino acids forming disulfide bridges in insulin to those not forming disulfide bridges in insulin.

(1)

Answer

(c) Suggest how insulin is synthesised from proinsulin.

Use the information in the diagram to support your answer.

(2)

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- (d) Insulin lowers the levels of glucose in the blood. This glucose is taken up by the liver and stored as glycogen.

Compare and contrast the structure of glucose with the structure of glycogen.

(3)

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*(e) Type 2 diabetes is a disorder that can result from ineffective or insufficient insulin. It is a risk factor for cardiovascular disease (CVD).

Cinnamon is a spice that has been shown to:

- increase the uptake of glucose from the blood into the liver
- increase the synthesis of glycogen.

In a study using rats, the effects of a cinnamon extract on levels of glucose in the blood was determined.

In this study a group of healthy rats and a group of rats with induced diabetes were used. The rats were given either the cinnamon extract or water, each day for 30 days.

The table shows the results of this study.

Time / days	Levels of glucose in the blood in healthy rats / mg dm^{-3}		Levels of glucose in the blood in rats with induced diabetes / mg dm^{-3}	
	Rats given water	Rats given cinnamon extract	Rats given water	Rats given cinnamon extract
0	106	102	395	378
10	101	103	403	346
20	99	96	419	384
30	94	90	493	369

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Explain the results of this study.

(6)

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(Total for Question 7 = 14 marks)

